

AI LinkedIn Post Generator using n8n & Groq AI

Project Report

Developed By

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1. Abstract

The AI LinkedIn Post Generator is an intelligent automation workflow developed using n8n and Groq AI. It automatically creates professional LinkedIn posts from a user-provided topic. The generated content includes a title, post, and hashtags. The workflow stores the generated data in Google Sheets and sends it via Gmail, reducing manual effort and improving productivity.

2. Introduction

Artificial Intelligence is transforming content creation by automating repetitive writing tasks. This project demonstrates how AI and workflow automation can be combined to generate professional LinkedIn posts. Using n8n, Groq AI, Google Sheets, and Gmail, the workflow automatically generates, stores, and emails LinkedIn content.

3. Problem Statement

Writing professional LinkedIn posts manually is time-consuming and requires creativity. This project automates the process, enabling users to generate high-quality LinkedIn content within seconds.

4. Objectives

- Automate LinkedIn post generation.
- Integrate Groq AI with n8n.
- Store generated content in Google Sheets.
- Automatically send the generated content via Gmail.
- Learn AI workflow automation.

5. Scope of the Project

This project is suitable for:

- Content Creators
- Digital Marketers
- Students
- Freelancers
- Professionals
- Social Media Managers

6. Software & Technologies Used

Technology	Purpose
n8n	Workflow Automation
Groq AI	Content Generation
JavaScript	JSON Parsing
Google Sheets	Data Storage
Gmail	Email Automation
Google OAuth	Authentication

7. System Requirements

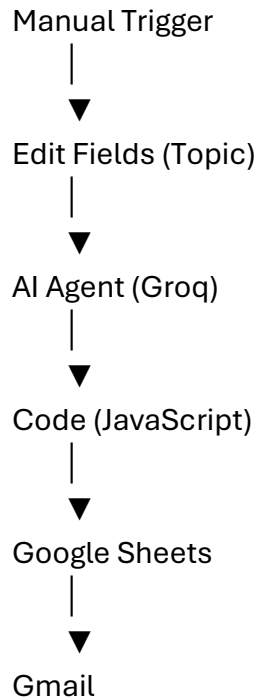
Hardware

- Intel i3 Processor or above
- 4GB RAM (Minimum)
- Internet Connection

Software

- Windows 10/11
- Google Chrome
- n8n
- Groq API Key
- Google Account

8. Workflow



9. Workflow Explanation

Step 1

The workflow starts with a Manual Trigger.

Step 2

The Edit Fields node accepts the topic entered by the user.

Step 3

The AI Agent sends the topic to Groq AI.

Step 4

Groq AI generates:

- Professional Title
- LinkedIn Post
- Hashtags

Step 5

The JavaScript Code node converts the AI response into structured JSON and adds the current date and time.

Step 6

Google Sheets stores the generated content.

Step 7

Gmail sends the generated LinkedIn post to the user's email.

10. Implementation Steps

- Install n8n
- Create Workflow
- Add Manual Trigger
- Add Edit Fields Node
- Configure AI Agent
- Connect Groq Chat Model
- Add JavaScript Code Node
- Configure Google Sheets
- Configure Gmail
- Execute Workflow

11. AI Prompt

You are a professional LinkedIn Content Writer.

Create a professional LinkedIn post on the following topic:

{{ \$json.topic }}

Instructions:

- Create a catchy and professional title.

- Write a LinkedIn post between 150 and 250 words.
- Use a professional and engaging tone.
- End the post with a question.
- Generate exactly 5 relevant hashtags.

Return ONLY valid JSON.

```
{  
  "title": "",  
  "post": "",  
  "hashtags": ""  
}
```

12. JavaScript Code

```
const text = $json.output.replace(/[\u0000-\u001F]+/g, "");
```

```
const data = JSON.parse(text);
```

```
const now = new Date();
```

```
return {  
  json: {  
    title: data.title,  
    post: data.post,  
    hashtags: data.hashtags,  
    date: now.toLocaleDateString("en-GB"),  
    time: now.toLocaleTimeString("en-US")  
  }  
};
```

13. Results

The workflow successfully generates:

- Professional Title
- LinkedIn Post

- Hashtags
- Date
- Time

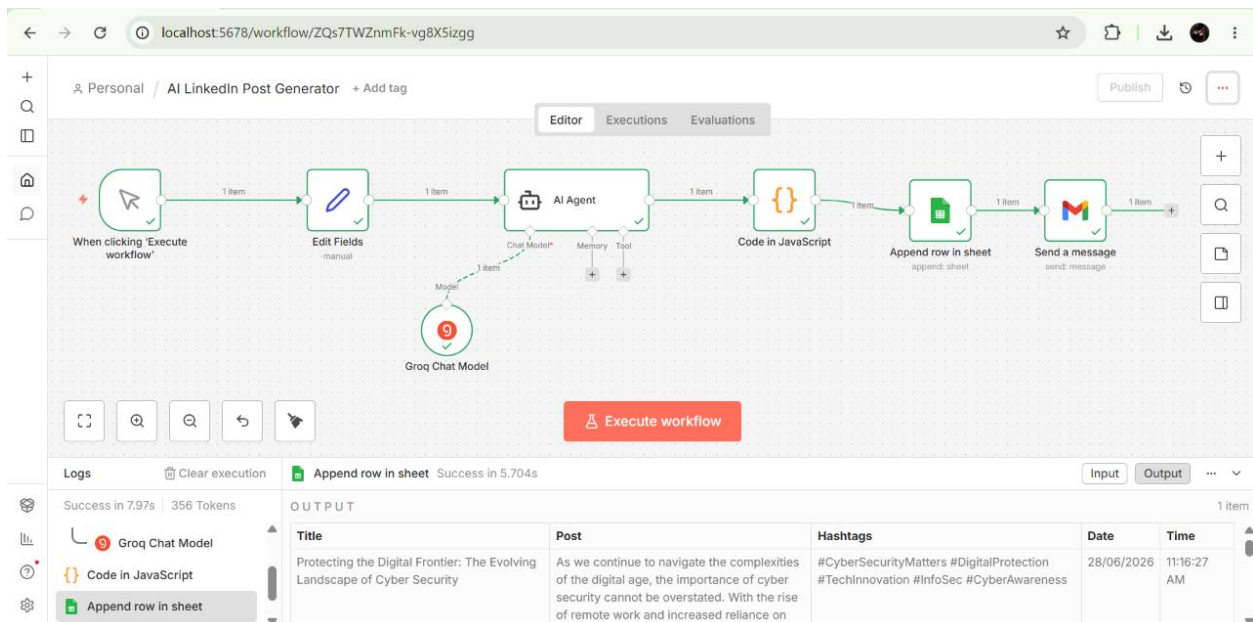
The generated content is automatically:

- Saved in Google Sheets
- Sent via Gmail

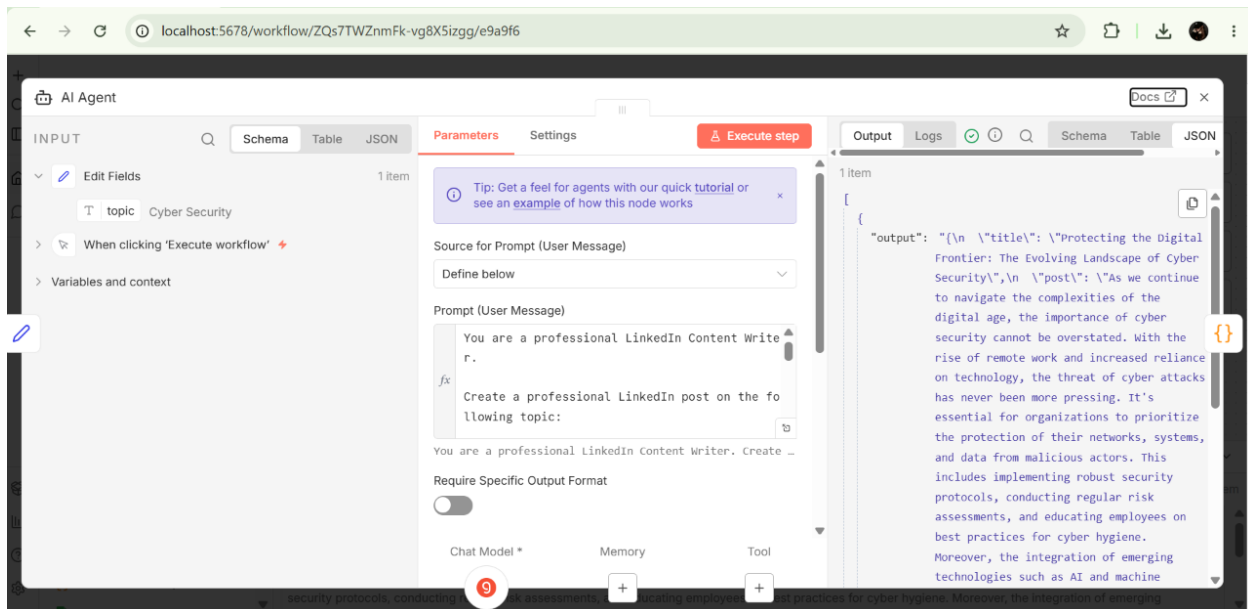
14. Screenshots

Add the following screenshots:

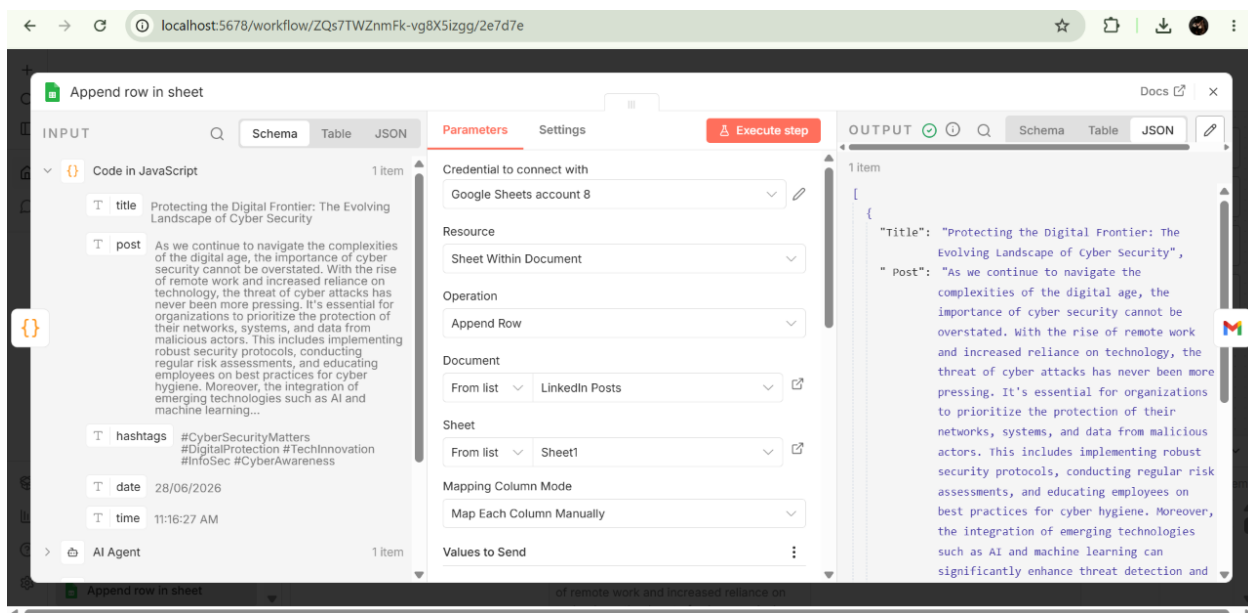
- Workflow



- AI Agent Output



- Google Sheets Output



- Gmail Output

The screenshot shows the 'Send a message' node configuration in a workflow editor. The node is titled 'Send a message' and has a 'Parameters' tab selected. The configuration includes:

- Credential to connect with:** Gmail account
- Resource:** Message
- Operation:** Send
- To:** nisarraheela024@gmail.com
- Subject:** Your LinkedIn Post
- Email Type:** Text
- Message:** (The message content is visible in the 'INPUT' tab)

The 'INPUT' tab shows a message with the following details:

- Title:** Protecting the Digital Frontier: The Evolving Landscape of Cyber Security
- Post:** As we continue to navigate the complexities of the digital age, the importance of cyber security cannot be overstated. With the rise of remote work and increased reliance on technology, the threat of cyber attacks has never been more pressing. It's essential for organizations to prioritize the protection of their networks, systems, and data from malicious actors. This includes implementing robust security protocols, conducting regular risk assessments, and educating employees on best practices for cyber hygiene. Moreover, the integration of emerging technologies such as AI and machine learning...
- Hashtags:** #CyberSecurityMatters #DigitalProtection #TechInnovation #InfoSec #CyberAwareness
- Date:** 28/06/2026
- Time:** 11:16:27 AM

The 'OUTPUT' tab shows the JSON response from the 'Send a message' node:

```
[{"id": "19f0ce9804b39f83", "threadId": "19f0ce9804b39f83", "labelIds": ["UNREAD", "SENT", "INBOX"]}]
```

A notification at the bottom right states: 'Node executed successfully'.

- Successful Execution

The screenshot shows the 'AI LinkedIn Post Generator' workflow diagram. The workflow consists of the following steps:

- When clicking 'Execute workflow'
- Edit Fields (manual)
- AI Agent (connected to Groq Chat Model)
- Code in JavaScript
- Append row in sheet (append: sheet)
- Send a message (send: message)

A notification at the bottom right states: 'Workflow executed successfully'.

The 'Logs' section shows the execution details for the 'Send a message' node:

Success in 7.037s | 362 Tokens

OUTPUT

id	threadid	labelids
19f0cea74223d4b5	19f0cea74223d4b5	0 : UNREAD 1 : SENT 2 : INBOX

A1	Title											
10	Revolutionizing ! The integration c #ArtificialIntelligence #HealthcareInnovation #MedicalTechnology #HealthcareTransformation #AlinHealthcare											27/06/2026
11	Revolutionizing ! The integration c #ArtificialIntelligence #HealthcareInnovation #DigitalHealth #MachineLearning #HealthcareTechnology #MedicalImaging #PredictiveAnalytics #Personalize											27/06/2026
12	Revolutionizing ! The integration c #ArtificialIntelligence #HealthcareInnovation #MedicalTechnology #AlinHealthcare #HealthcareTransformation #DigitalHealth #InnovationInH											27/06/2026
13	Revolutionizing ! The integration c #ArtificialIntelligence #HealthcareInnovation #MedicalTechnology #AlinHealthcare #HealthcareTransformation #DigitalHealth #Innovation											27/06/2026
14	Elevating Your B As we continue t #DigitalMarketing #MarketingStrategy #BrandAwareness #SocialMedia #ContentMarketing #SEO #EmailMarketing #LeadGeneration											27/06/2026
15	Elevating Your Brand with Digital Marketing											
16	Strengthening C: As we continue t #CyberSecurity #DigitalTransformation #InfoSec #SecurityAwareness #Technology											27/06/2026
17	The Evolving Lai As we continue t #CyberSecurity #InfoSec #DigitalProtection #SecurityMeasures #DataProtection #StaySafeOnline											28/06/2026
18	Protecting the Di As we continue t #CyberSecurityMatters #DigitalProtection #InfoSec #CyberAwareness #TechInnovation											28/06/2026
19	Protecting the Di As we continue t #CyberSecurityMatters #DigitalProtection #InfoSec #CyberAwareness #TechInnovation											28/06/2026
20	Protecting the Di As we continue t #CyberSecurityMatters #DigitalProtection #InfoSec #CyberThreatIntelligence #SecurityAwareness											28/06/2026
21	Protecting the Di As we continue t #CyberSecurityMatters #DigitalProtection #TechInnovation #InfoSec #CyberAwareness											28/06/2026
22	Protecting the Di As we continue t #CyberSecurityMatters #DigitalProtection #TechInnovation #InfoSec #CyberAwareness											28/06/2026
23	Strengthening C: As we continue t #CyberSecurityMatters #DigitalProtection #OnlineSafety #TechSecurity #CyberAwareness											28/06/2026
24												
25												
26												

15. Challenges Faced

- Groq API configuration
- JSON parsing
- Gmail OAuth authentication
- Google Sheets field mapping
- Undefined values
- AI prompt optimization

16. Solutions

- Used Groq Chat Model
- Added JavaScript Code node
- Configured Gmail OAuth
- Correctly mapped Google Sheets columns
- Improved AI prompt

17. Learning Outcomes

This project helped in learning:

- AI Workflow Automation
- n8n
- Prompt Engineering
- Groq AI
- JavaScript
- Google Sheets Integration
- Gmail Automation
- API Integration
- Workflow Debugging

18. Future Enhancements

- Add Web Form for topic input
- Publish directly to LinkedIn
- Generate AI images
- Schedule automatic posts
- Support multiple social media platforms

19. Conclusion

The AI LinkedIn Post Generator successfully demonstrates the integration of Artificial Intelligence with workflow automation. It automates content creation, data storage, and email delivery, making it a practical solution for professionals and content creators while showcasing modern AI automation skills.

20. References

- n8n Documentation
- Groq API Documentation
- Google Sheets API Documentation
- Gmail API Documentation

- JavaScript Documentation